1KOSMOS

AR Master Sheet and ARR Dashboard Correlation Analysis

Analysis of **AR Dashboard - Master Sheet - 15 June.xlsx** and **1Kosmos — ARR Dashboard.html**, including common dimensions, missing transaction keys, business meaning, and recommended integration logic.

Document: 1Kosmos AR–ARR Correlation Analysis
AR source: AR Dashboard - Master Sheet - 15 June.xlsx
ARR source: 1Kosmos — ARR Dashboard.html
Assessment date: 22 June 2026

1. Executive Conclusion

Correlation exists, but the two files are not directly connected.
The ARR HTML is generated from Booking Database-style contract and monthly ARR data. It does not load the AR workbook and does not contain invoice, due-date, open-balance, or payment-event fields. However, both files describe many of the same customers, regions, and sales representatives. They can therefore be integrated at customer/account level after names are standardized.

The files answer different questions:

ARR dashboard	AR master sheet
How much recurring revenue is contracted?	How much invoiced money is unpaid?
How ARR changes through new, upsell, downsell, churn, and renewal events.	Which invoices are open, overdue, paid, or pending renewal/invoicing.
Commercial and revenue-retention intelligence.	Receivables, collection, and cash-risk intelligence.

2. Files Analyzed

2.1 AR Dashboard Master Sheet

The workbook contains three functional sheets:

Sheet	Purpose	Observed data
AR Aging	Current unpaid invoices and balances.	30 rows; customer, end user, region, sales rep, document date, invoice number, due date, open balance.
Payment History	Invoice and payment movements by invoice number.	340 meaningful rows; 150 invoice numbers; invoice entries are positive and payment entries are negative.
Renewal Pending	Commercial renewals or signed deals pending action.	5 rows; end user, pending period/status, amount, sales rep, region, remarks.

2.2 ARR Dashboard HTML

The HTML contains an embedded JavaScript object named `D`. It has 303 deals, 90 monthly periods, 80 unique end users, 21 sales representatives, and regions APAC, NAM, and OEM. Its latest embedded month is June 2026.

The deal structure contains fields such as:

```
key_id, end_user, region, sales_person, size, industry,
sub_product, booking_type, mode, contract_id, contract_name,
order_status, rec_non_rec, term_start, term_end, arr, changes
```

The HTML also contains aggregate ARR fields including current ARR, previous ARR, monthly ARR, monthly changes, new ARR, upsell, churn, downsell, GRR, NRR, active logos, regions, products, and sales-person analysis.

3. AR Workbook Summary

Open AR balance	AR Aging rows	Payment-history invoices
$10.77M	**30**	**150**

Payment-history rows	Pending renewal amount	Pending renewal rows
340	**$1.24M**	**5**

Using 15 June 2026 as the workbook's analysis date, the open-balance aging profile is:

Aging bucket	Open balance
Not yet due	$2,879,046.85
0–30 days overdue	$3,410,000.00
31–60 days overdue	$298,273.72
61–90 days overdue	$170,000.00
90+ days overdue	$4,011,980.00

The largest observed open balances include Lowes ($4.01M), Concentrix Solutions Corporation ($3.41M), DMDC ($810K), Staples entries, and S&P Global.

4. Measured Correlation

AR sheet	Customer/end-user overlap with ARR HTML	Sales-rep overlap	Interpretation
AR Aging	19 of 21 normalized customer names (about 90.5%)	5 of 7 normalized rep names	Strong account-level relationship.
Payment History	37 of 37 normalized customer names (100%)	7 of 8 normalized rep names	Very strong historical account overlap.
Renewal Pending	2 of 5 direct normalized names	1 of 2 normalized rep names	Partial overlap; several values are aliases or commercial labels such as Rockwell_Upsell and Kotak.

Examples of shared customers include Convera, Great Wolf Lodge, Vodafone, Kforce, Kotak Mahindra Bank, Lowes, Concentrix Solutions Corporation, Rockwell Automation, S&P Global, Staples, DMDC, VF Services, Hitachi, Netsmart, University of Michigan, and The Auto Club Group.

Name standardization was required. Examples include Rockwell Automation, Inc. → Rockwell Automation; Staples, Inc. → Staples; Concentrix → Concentrix Solutions Corporation; and Salesforce, Inc. → Salesforce.

5. Direct Field Mapping

AR field	ARR field	Mapping quality	Comment
End User	end_user	Strong after normalization	Best common business dimension.
Region	region	Strong	NAM and APAC align directly. OEM may not appear in AR.
Sales Rep	sales_person	Strong with aliases	Some names or team labels need a mapping table.
Customer / Customer Name	No exact bill-to field in embedded deal object	Moderate	ARR end user can be used as fallback, but legal bill-to and end user are not always identical.
Renewal amount	ARR changes / renewal context	Business correlation only	Cannot be joined transaction-by-transaction without a contract identifier.

6. Fields with No Direct Match

AR-only field	Why it matters	ARR HTML status
Invoice number / document number	Provides an exact receivable transaction key.	Not present.
Invoice date	Identifies billing timing.	Not present.
Due date	Required for aging and overdue calculations.	Not present.
Open balance	Measures unpaid receivables.	Not present.
Payment type and payment amount	Supports collection and days-to-pay analysis.	Not present.
Renewal status and remarks	Explains pending commercial actions.	Not present as structured source fields.

No exact transaction join currently exists. The ARR HTML has Contract ID but the AR workbook has Invoice Number. Neither file supplies an invoice-to-contract bridge. Matching only by customer can combine unrelated contracts for the same account.

7. Meaningful Business Correlations

7.1 ARR versus open receivables

Joining by a canonical customer identifier allows the dashboard to show contracted recurring revenue alongside unpaid balance:

```
AR exposure ratio = Open AR balance / Current ARR
```

A high ratio can identify accounts where a large part of annual recurring revenue is exposed to collection risk.

7.2 Overdue AR as churn or renewal risk

Overdue invoices do not automatically cause churn, but they provide a useful risk signal. Customers with high ARR and severely overdue balances can be highlighted for Finance, Customer Success, and Sales coordination.

7.3 Payment behavior and customer health

Payment History can produce average days to pay, late-payment frequency, outstanding invoice count, and collection trend. These can enrich Customer 360 and expansion-risk models in the ARR dashboard.

7.4 Renewal Pending and ARR forecast

The Renewal Pending sheet is commercially closer to ARR than the other AR sheets. Pending annual renewals, multi-year installments, and signed-but-not-invoiced deals can enrich forecast and at-risk views. A contract-level key is still needed for reliable automation.

8. Recommended Integrated Data Model

```
Booking Database → ARR Contract / Monthly ARR tables
AR Aging → Open Invoice table
Payment History → Invoice Payment Event table
Renewal Pending → Renewal Risk table
All tables → Canonical Customer + Sales Rep + Region dimensions
Canonical dimensions → Combined ARR and AR Dashboard APIs
```

Recommended table	Key fields
Customer	customer_id, canonical_name, legal_name, end_user_name, aliases
ARR Contract	contract_id, customer_id, product, region, sales_rep, monthly_arr
AR Invoice	invoice_number, customer_id, invoice_date, due_date, original_amount, open_balance
Payment Event	invoice_number, event_date, event_type, amount
Renewal Risk	customer_id, contract_id when available, status, pending_amount, remarks

Recommended table	Key fields
Customer Alias	source_system, source_name, customer_id

9. Required Join Strategy

1. Create a permanent canonical customer ID.
2. Maintain aliases such as Staples, Staples Inc., and legal bill-to names.
3. Map sales-team aliases such as CS Team to responsible owners when required.
4. Add Contract ID or customer master ID to the AR workbook.
5. Prefer Invoice Number → Contract ID mapping when one invoice relates to a specific contract.
6. Do not rely on fuzzy customer names as the final production join.

Suggested temporary POC matching priority:

```
1. Exact canonical customer ID
2. Exact normalized end-user name
3. Approved customer-alias mapping
4. Manual review
5. Never auto-join on fuzzy similarity alone
```

10. Proposed Combined Dashboard KPIs

KPI	Calculation
Total ARR	Sum of latest monthly ARR.
Total Open AR	Sum of open invoice balances.
Overdue AR	Open balance where due date is before the selected as-of date.
90+ Day AR	Open balance overdue by more than 90 days.
AR Exposure Ratio	Total Open AR ÷ Total ARR.
High-Risk ARR	ARR from customers with severe overdue balances or renewal issues.
Average Days to Pay	Average payment date minus invoice date or due date, depending on policy.
Pending Renewal Value	Sum of Renewal Pending amount.
Signed but Not Invoiced	Renewal rows with that status.

11. Suggested Combined Customer 360

For each customer, a combined view could display:

- Current ARR and monthly ARR movement.
- Products, contracts, region, and sales owner.

- Total open balance and overdue aging buckets.
- Open invoices and payment history.
- Average payment delay and recent late-payment pattern.
- Pending renewal status, amount, and remarks.
- Risk label such as Healthy, Watch, High Risk, or Action Required.

12. Final Assessment

The two files are highly related from a customer and account-management perspective, but they represent different financial processes:

- **ARR** measures recurring revenue value and movement.
- **AR** measures invoicing, unpaid balances, and payment collection.

The observed customer overlap is strong enough to support a combined POC. Production integration should add canonical customer IDs and an invoice-to-contract bridge. Until those identifiers exist, AR and ARR should be correlated at customer level and clearly labeled as an account-level relationship rather than an exact contract-level reconciliation.

Generated from the supplied AR master workbook and ARR dashboard HTML on 22 June 2026.